STATISTICS

Chapter 4 Homework

Discrete random variables and model choice

Sections 4.1-4.5 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Explain your reasoning in words. Use the context to identify what is counted, how trials or draws are conducted, and which assumptions support each model. All scenarios are fictional.

Read and interpret a discrete distribution

1. A delivery company tracks the number of late packages in a shift and the exact time until its next late delivery. Which variable is discrete? Explain how the possible values of the two variables differ.

2. One proposed probability table lists all possible outcomes and has a total probability of one, but an entry is negative. Another lists every possible outcome and has no negative entries, but its probabilities total less than one. Explain why each table is invalid.

3. A manager reads a probability table for the number of late packages. Explain which rows belong to “at least one is late” and how that event relates to “none is late.”

4. A model for late packages gives a long-run expected count between two whole numbers. Explain why that makes sense and what you could learn from the standard deviation that you cannot learn from the expected count alone.

5. Two games have the same expected net gain. One pays the same amount every time, while the other sometimes pays more and sometimes less. Explain how the games can have the same expected value but different risk.

Choose and compare models

6. An inspector checks a fixed number of packages independently. Each package has the same chance of being damaged. Identify the distribution for the number of damaged packages and explain what makes it a good fit.

7. A caller makes independent attempts, each with the same chance of reaching a person. Identify the distribution for the attempt on which someone first answers. Explain whether that successful attempt is included in the count.

8. A teacher randomly selects a fixed-size sample of cards from a mixed stack of marked and unmarked cards without replacing them. Identify the distribution for the number of marked cards and explain what happens to the chance of drawing a marked card after one is removed.

9. Now suppose the teacher returns each card and mixes the stack before drawing again. The teacher still makes a fixed number of draws. Which model now fits the marked-card count, and what does the replacement change?

10. Two machines have independent attempts with a constant success chance for each machine. One has a higher success chance than the other. Compare their typical waits for a first success and explain whether either has a guaranteed finite maximum wait. Assume both chances are strictly between zero and one.

11. Explain how to distinguish binomial, geometric, and hypergeometric situations by asking what the variable counts, when the procedure stops, and whether sampling changes the pool.